



Society of Petroleum Engineers

SPE-229603-MS

Enhancing Risk Analysis in Drilling Operations Using an Agentic LLM Workflow

P. Chica, Agustin Soriano Rementeria, A. Hussein, and B. El Koussa, SLB

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229603-MS](https://doi.org/10.2118/229603-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper presents a novel, modular agentic workflow leveraging large language models (LLMs)—a technology increasingly validated for its ability to handle complex textual data in drilling operations (e.g., [Ferrigno et al., 2024](#))—and deterministic rule-based heuristics to automate high-fidelity extraction and classification of drilling risk events from daily reports. The system translates unstructured text into structured risk objects, enabling seamless integration with enterprise databases and significantly reducing manual risk analysis workload. The solution was developed as a rapid proof of concept (POC) within eight weeks, designed to pragmatically validate the approach with open-source datasets and scalable architecture. Validation on open-source datasets demonstrated nearly 90% recall for risk event detection and 75% categorization accuracy. The workflow surfaced more than 30 hours of previously unrecognized diminished capacity within productive time (PT) intervals for a single well, while reducing manual review time from 3 hours to less than 5 minutes per well. By surfacing these hidden risks and enabling fine-grained classification, the workflow supports more proactive, data-driven decision-making for drilling engineers and operational teams. This method establishes a new benchmark for automated, consistent, and scalable risk analysis across a broad spectrum of drilling operations.

Introduction and Industry Context

In recent years, the oil and gas industry has witnessed rapid advancements in digitalization and artificial intelligence, enabling greater operational efficiency, risk mitigation, and data-driven decision-making. Despite substantial technological advances, a significant gap remains in the systematic detection and documentation of operational hazards—especially those events recorded in Daily Drilling Reports (DDRs) that are buried within productive-time intervals and unstructured notes, leaving them inaccessible to efficient analysis. Manual interpretation of daily drilling reports (DDRs), even with modern data solutions, often struggles to reveal subtle inefficiencies and risks that gradually degrade performance without generating clear non-productive time (NPT) records. These challenges stem from the labor-intensive, error-prone, and inconsistent nature of manually reviewing vast amounts of free-text comments, which can take hours and are subject to human error and subjective content, particularly from operators across different regions ([Ferrigno et al., 2024](#); [Wang et al., 2025](#)).

Large language models (LLMs) and generative AI (GenAI) technologies are now transforming how textual and semi-structured drilling data can be analyzed, overcoming previous limitations of keyword matching or basic NLP that struggled with context, ambiguity, and domain-specific language. Recent works, such as [Ferrigno et al. \(2024\)](#) and [Wang et al. \(2025\)](#), have shown that agentic workflows powered by LLMs can automate the classification and extraction of critical drilling events, significantly reducing the need for labor-intensive manual reviews and boosting efficiency by orders of magnitude.

For instance, [Ferrigno et al. \(2024\)](#) demonstrated the revolutionary impact of integrating LLM-based AI models into drilling control rooms, enhancing decision-making efficiency by over 50 times and reducing search and background analysis time from hours to minutes through the accurate, zero-shot classification of complex WITSML data into 42 distinct categories. Similarly, [Wang et al. \(2025\)](#) presented an innovative GenAI framework for automating daily drilling mud report processing, achieving near-perfect accuracy, reducing manual effort by 99%, and enabling robust business intelligence dashboards. Furthermore, [Amour et al. \(2024\)](#) demonstrate the potential of natural language-to-SQL (NL2SQL) GenAI frameworks to unlock rapid and nontechnical access to complex drilling data, bridging the gap for engineers lacking specialized data skills. These developments establish a clear precedent for the pragmatic application of LLM-based automation in drilling operations, aligning with broader industry goals of safety, operational excellence, and continuous improvement.

This paper presents an agentic LLM workflow for automated risk event extraction and classification from daily drilling records, specifically designed to reveal risks hidden within PT intervals. Notably, this solution was conceived and executed as a rapid proof of concept (POC) over just eight weeks, leveraging open-source datasets and primarily using GPT-4 and its variants. The approach emphasizes modularity, auditability, and scalability, with the aim of confirming that high-fidelity risk detection and classification can be achieved in a practical, repeatable manner.

Project Overview and Methods

The project originated from the recognition that a large fraction of operational inefficiencies and risk events in drilling are overlooked or misclassified in standard reporting workflows. Most risk classification remains binary, and risk events that degrade performance but do not trigger NPT often pass unnoticed. The goal of this work was to design, implement, and field-validate a modular, agentic workflow based on large language models (LLMs) for high-fidelity extraction and classification of drilling risk events embedded in daily operations records.

Proof of Concept Approach

The entire solution was conceived and delivered as a pragmatic, time-boxed proof of concept (POC), completed in just eight weeks. This POC structure mirrored the rapid, iterative methodology recommended in recent industry research ([Ferrigno et al., 2024](#)), emphasizing early stakeholder feedback, real-world validation, and scalability. The system was implemented using a combination of open-source data (Utah FORGE, LORIS), allowing benchmarking against both published literature and internal standards.

Workflow Architecture

The core of the workflow is a modular agentic pipeline in which LLM "agents" are guided by deterministic rules to produce structured outputs from unstructured text. Each drilling comment is processed in multiple steps:

1. **Risk Event Detection:** The first agent classifies the comment as either a risk event or not, leveraging prompt engineering and context-aware instructions.

2. **Domain Assignment:** For comments flagged as risk, a specialized agent assigns the primary risk domain (e.g., equipment failure, wellbore stability, hydraulics, or mechanical).
3. **Subcategory Classification:** Domain-specific agents classify the event into one of 12 operational subcategories, ensuring logical consistency and operational realism.

The workflow employs schema-driven output protocols and rule-based logic to ensure deterministic mapping from free text to risk objects. Key challenges addressed include resolving linguistic ambiguity, ensuring domain adaptation, and managing data sparsity—issues common to many AI applications in the energy sector.

Data Preparation and Labeling

A ground-truth dataset of over 1,200 drilling comments was created, spanning onshore and offshore wells and diverse operational contexts. Comments were manually labeled by subject matter experts (SMEs) using a consensus-based, double-pass validation protocol. Special attention was given to the "gray zone" cases—comments that could plausibly fit more than one risk category—by requiring secondary review and team consensus.

Example comment labeling

"Minor seepage losses noted while drilling ahead" Labeled as:

- Risk Hydraulics Loss Circulation.
- Risk Equipment failure Surface Systems

Iterative Model Development

The journey to an effective AI-driven risk analysis system began with early testing of single-agent Large Language Models (LLMs). These initial attempts quickly highlighted a significant problem: the models achieved sub-60% accuracy, a performance level clearly inadequate for reliable operational insights. This result underscored the critical need for a more sophisticated approach, specifically one that embraced specialized, context-aware prompt engineering. The challenge came from the start when in one single pass, through one prompt, it was expected to have a complex, multi-layered classification model. This meant the model had to simultaneously identify whether a drilling log entry represented "no risk" or an "undesired event," then assign it to one of five broad risk domains (like equipment failure or wellbore stability), and finally, pinpoint one of twelve detailed subcategories, all in one go. This monolithic approach, while conceptually straightforward, proved to be overly simplistic for a wide range and often ambiguous nature of real-world drilling comments. It was like asking a generalist to be an expert in a dozen fields at once; the depth of understanding required simply wasn't there.

Recognizing these limitations, the development shifted towards an iterative and modular strategy, transforming the workflow into a multi-stage agentic pipeline as showed in [Fig 1](#). This evolution was pivotal. Instead of a single, overburdened LLM, the system now comprises specialized agents, each designed to tackle a specific part of the classification problem. Each of these agents receives finely tuned, structured prompts that are meticulously crafted to guide its understanding. Furthermore, they are fed contextual cues and a rich array of edge-case examples, enabling them to navigate the complexities and subtleties inherent in drilling narratives. This rigorous refinement process was absolutely crucial for significantly improving both the accuracy and robustness of the entire system.

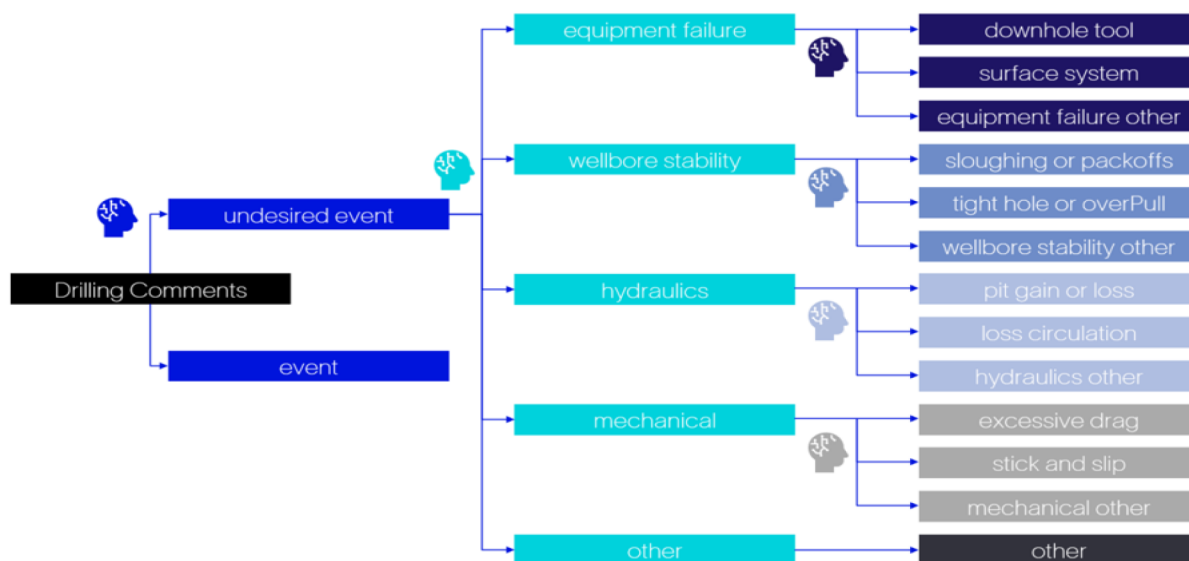


Figure 1—Three layers to classify risk through LLM agents

A key learning that emerged during these intensive iterations was the frequent occurrence of multiple distinct risks within a single comment. For example, a single log entry might describe both a minor gain and a downhole failure, indicating two different types of operational risks. Initially, the single-pass model often struggled with these compound entries, arbitrarily assigning only one risk type, which might not even be the most relevant or impactful one. To meticulously address this complex challenge and ensure comprehensive risk capture, a specific order of precedence for classification was established for comments where multiple risks were clearly present. This hierarchy prioritizes: wellbore stability followed by hydraulics, then mechanical issues, equipment failure, and finally, a catch-all "others" category for any legitimate risks that didn't fit the primary defined domains. This structured approach ensured that all identified risks were categorized systematically and appropriately.

To further bolster the system's reliability and to minimize the notorious problem of LLM hallucinations—where models generate plausible but incorrect or fabricated information—a rule-based post-processing layer was seamlessly integrated. This critical layer acts as a final safeguard, enforcing strict alignment between the assigned subcategories and their designated parent domains. It provides a deterministic handling mechanism for those rare or particularly ambiguous cases that might still challenge the LLM agents, ensuring that even in uncertain scenarios, the classification adheres to predefined logical constraints. The overall performance of this refined system was rigorously validated against a diverse set of data, including both manually labeled datasets (our ground truth) and unseen holdout datasets sourced from entirely new wells. This comprehensive validation process was vital for ensuring the system's generalizability and scalability across a broad spectrum of real-world operational scenarios in the drilling industry.

System Integration

The development of this risk analysis solution did not stop at merely identifying hidden inefficiencies; its true power lies in its seamless integration into the broader operational ecosystem. The finalized workflow was meticulously designed for deep integration with enterprise SQL databases. This wasn't just about storing data; it was a deliberate architectural choice made to significantly boost the existing capabilities of the enterprise system. By translating unstructured daily drilling comments into structured risk objects, and critically, mapping all outputs directly to WITSML risk object schemas, the solution ensures seamless data flow and interoperability with other enterprise applications.

This integration provides immediate, tangible benefits. It enables direct visualization of previously unseen risks, allowing engineers and operational teams to quickly grasp the severity and patterns of diminished capacity. Furthermore, it empowers them to slice and dice this newly surfaced risk data, performing in-depth analytics to understand the precise operations performed at diminished capacity as described in Fig 2. This granular insight into performance bottlenecks, which traditional Non-Productive Time (NPT) tracking often misses, significantly enhances the enterprise system's ability to provide a comprehensive operational picture. It transforms raw, qualitative observations into quantifiable, actionable intelligence that can be integrated with other operational data for a holistic view of well performance.

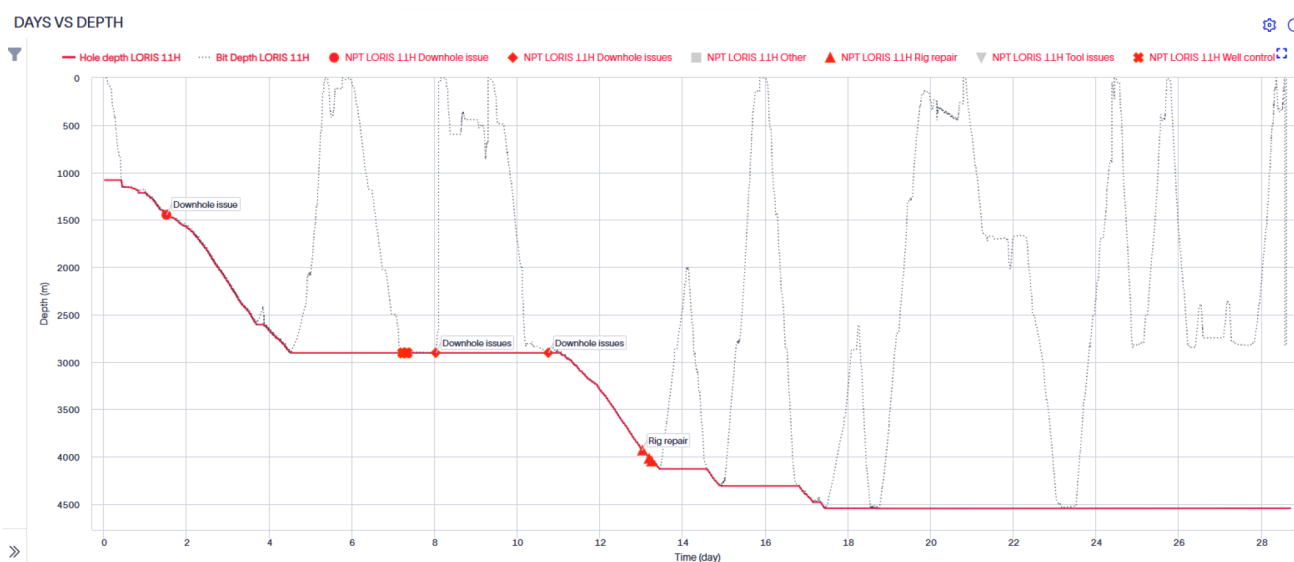


Figure 2—Integration of events in days vs depth in Enterprise visualizer

Beyond initial deployment, the solution incorporates a closed-loop architecture, which is fundamental for its long-term viability and accuracy. This design isn't static; it enables continuous improvement through direct feedback from Subject Matter Experts (SMEs) and iterative prompt logic updates. This practice is now widely recognized as essential for sustainable, high-quality Generative AI (GenAI) deployments, especially in dynamic environments like drilling. By continuously learning from new data and expert input, the system becomes more intelligent and precise over time, ensuring that the enterprise system consistently benefits from the most accurate and nuanced risk insights available. The integration thus extends beyond data exchange, fostering an adaptive and evolving system that continuously refines its understanding of operational risk.

Results

Performance and Validation

The agentic workflow was rigorously benchmarked on multiple LLM engines, including GPT-4o and its variants (GPT-4.1, GPT-4.1 mini). Through this evaluation, GPT-4o ultimately provided the best tradeoff between speed, cost, and accuracy for this specific application. Key metrics from field validation include:

- Risk Event Detection Recall: >90%
- Domain and Subcategory Accuracy: ~75%
- False Positive Rate: <10%, primarily observed in ambiguous or poorly documented cases
- Manual Review Time Reduction: From approximately 3 hours to under 5 minutes per Well

These results demonstrate significant operational efficiencies. Notably, the observed reduction in manual review time per well, achieving a greater than 97% reduction (from 3 hours to less than 5 minutes), is highly competitive with, and in some aspects surpasses, the efficiency gains reported in other recent industry deployments of LLM-based solutions for similar analytical tasks. This substantial improvement reinforces the practical, rapid value and transformative potential that LLM-driven automation brings to complex, unstructured drilling data analysis.

Operational Impact

The workflow uncovered over 30 hours of previously unrecognized diminished capacity in PT intervals for a single well. The performance of the risk agentic model showed >90% recall. All results were returned to the SQL database for immediate analytics and operational review.

Table 1—Integration of events in days vs depth in Enterprise visualizer

Example	Output			Table:
	Comment Text	Binary Layer	Risk Domain	Subcategory
	Directionally drilled (8 1/2") from 4,394m to 4,435m. Seepage losses of up to 3.18m3 (20bbl) per hour while drilling	Undesired	Hydraulics	Loss Circulation
	Rotated string down due to drag from 4,106m to 4,251m with 757Lpm (200gpm), 3.72 to 4.0MPa (540 to 580psi), 30rpm and 13.6 to 19kNm (10 to 14kft.lbs) torque	Undesired	mechanical	excessive drag
	Clean up cycle w/twohigh vis sweeps.Spot high vis pill on bottom.	Regular	N/A	N/A
	ROP including downlinking and connections: 28.6m/hr. Reamed stand up and down prior to connections. Stickslip impacted directional control. Reduced rotary speed to 120rpm to make directional corrections. Directionally drilled 311mm (12 1/4") hole from 2,232m to 2,404m.	Undesired	mechanical	stick and slip

Discussion

Key Challenges and Lessons Learned

The development of an AI-driven system for advanced risk analysis in drilling operations revealed several important challenges and lessons. One primary challenge was the inconsistency of language found in daily drilling reports. Comments from field personnel, written under various operational context and circumstances, frequently contain specialized terms, abbreviations, and informal expressions. This diversity in terminology and narrative style makes automated interpretation of events complex. To address these issues, a multilayered approach was implemented. Specialized large language model (LLM) agents progressively refined text classification, with prompt instructions and examples carefully crafted to improve contextual understanding. In addition, rule-based logic was integrated as a deterministic safeguard, providing reliable classification even when subtle variations or ambiguities were present in the text. This hybrid approach has proven effective for dealing with the variability of unstructured field notes, delivering classifications more accurate results.

Subject Matter Expert (SME) engagement emerged as another essential component of system development. The quality of ground-truth data, used to train and validate the models, directly influenced

overall performance. SME contributions extended beyond data labeling, including input on prompt logic and the differentiation of closely related operational events. Ongoing feedback from SMEs was especially valuable for handling ambiguous entries and validating edge cases, ensuring that model outputs matched practical, real-world understanding. This collaborative, human-in-the-loop strategy underscores the ongoing necessity of domain expertise for establishing robust and reliable automation in complex environments.

Adaptability to different operational environments was also a key requirement. The workflow demonstrated strong performance across diverse settings, effectively classifying risks from both land-based and offshore operations. This adaptability results from continual prompt and data updates, enabling the system to incorporate new examples, regional terminology, and evolving operational practices. The iterative update process ensures ongoing relevance and accuracy as new data becomes available.

The continuous improvement process was also identified as fundamental to long-term effectiveness. The system architecture was designed to support ongoing updates and expansion, accommodating input from a growing range of operational fields and geographies. This approach facilitates the rapid identification and integration of new or unforeseen edge cases into model training and prompt design. The ongoing refinement process aims to further improve model accuracy, particularly in categorizing risks within four main groups—hydraulics, wellbore stability, mechanical, and equipment failure—as well as twelve detailed subcategories. Comprehensive risk coverage is also maintained by including a fifth "other" category, ensuring that outliers and atypical cases are not overlooked.

Limitations and Future Directions

Despite the promising results of this proof of concept, several areas for future development remain. Expanding the classification framework to cover a wider range of operational events—including specific health, safety, and environmental incidents, specific operations like cementing or liner hanger risks, or quality deviations will increase the system's value. Integration with real-time data sources may enable proactive risk identification, moving from retrospective analysis to on-the-fly decision support.

The growing international scope of drilling operations highlights the need for multilingual model support. Further advances could extend the system's capabilities from risk detection to automated recommendations, including suggested operational responses or references to standard procedures, thereby enhancing its function as a comprehensive decision-support tool.

Conclusions

The integration of large language models (LLMs) with deterministic, rule-based logic in a modular, agentic workflow has proven to be a powerful enabler for robust, scalable, and explainable risk classification in complex drilling operations. This work provides tangible evidence that advanced AI can be effectively harnessed to tackle the persistent challenges of extracting actionable intelligence from unstructured daily drilling reports—documents that traditionally required significant manual review, suffered from inconsistency, and left critical operational insights buried within narrative text.

Developed and delivered within an intensive eight-week proof-of-concept period, the agentic workflow successfully bridged the gap between free-text operational narratives and structured, actionable risk intelligence. The results clearly demonstrate that the combination of LLMs and rule-based heuristics does not merely automate existing processes, but actively enhances the fidelity, speed, and transparency of risk detection and classification. The workflow's deployment resulted in the identification of over 30 hours of previously unrecognized diminished-capacity intervals within productive time for a single well, an achievement that would have been nearly impossible through manual methods alone. Manual analysis time was reduced by more than 90%, compressing hours of tedious review into just minutes—thereby

enabling engineers and operational teams to devote their expertise to higher-value analysis, intervention, and optimization.

Importantly, the modularity and transparency of this approach offer significant advantages over "black-box" machine learning models. The workflow's explainability is rooted in its layered, agent-based structure, where each stage of classification is not only auditable but also easily updated or refined based on feedback from subject matter experts (SMEs). This ensures that the system remains adaptable to evolving operational contexts, regional linguistic nuances, and new types of drilling challenges. The closed-loop architecture supports continuous improvement, allowing the incorporation of new data, updated prompts, and ongoing SME input—a practice essential for long-term success and model accuracy in rapidly changing operational environments.

In summary, this project not only validates the viability of LLM-driven, agentic workflows for risk analysis in drilling, but also establishes a foundation for broader adoption of explainable AI solutions in industrial domains where accuracy, transparency, and operational impact are paramount.

References

1. Ferrigno, E., Rodriguez, M., & Davidsson, E. (2024). Revolutionizing Drilling Operations: Next-gen LLM- AI for Real-time Support in Well Construction Control Rooms. SPE-220798-MS.
2. Wang, J., Yoon, J., Marzban, A., Castanos, R., Fruge, N., & Holman, I. (2025). Automatic Daily Drilling Mud Report Processing Using Generative AI to Maximize the Operational Efficiency. OTC-35625-MS.
3. Amour, M., Rachmat, B. K., Rementeria, A. S., & Guillot, V. (2024). Empowering Drilling and Optimization with Generative AI. SPE-221862-MS.